

120 Days of Quant Finance

Week 4 – Financial Theory

Day 6: Options Use Cases and Risk Metrics

Arnav Garg

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Introduction

Options play a central role in modern financial markets, not primarily as speculative instruments but as tools for hedging, income generation, volatility trading, and risk management. Evaluating option-based strategies requires risk metrics that go beyond simple return and variance.

This section discusses real-world option use cases and introduces key performance and risk measures used in practice.

Options Use Cases in Real Markets

Hedging and Portfolio Insurance

Protective options are commonly used to hedge downside risk. By purchasing put options, investors can limit losses while maintaining upside exposure, at the cost of an upfront premium.

Income Generation

Covered call strategies generate income by selling call options against an existing asset position. This approach sacrifices some upside potential in exchange for steady option premium income.

Volatility Trading

Many option strategies are designed to trade volatility rather than direction. Examples include straddles and strangles, which profit from large price moves regardless of direction.

Tail-Risk Management

Deep out-of-the-money options are used to protect against rare but extreme market events. These strategies are commonly employed by institutional risk managers.

Risk and Performance Metrics

Sharpe Ratio

The Sharpe ratio measures excess return per unit of total volatility:

$$\text{Sharpe Ratio} = \frac{E[R_p - R_f]}{\sigma_p},$$

where R_p is portfolio return, R_f is the risk-free rate, and σ_p is portfolio volatility.

Sortino Ratio

The Sortino ratio focuses on downside risk by replacing total volatility with downside deviation:

$$\text{Sortino Ratio} = \frac{E[R_p - R_f]}{\sigma_{\text{down}}}.$$

Maximum Drawdown

Maximum drawdown measures the largest peak-to-trough loss over a given period:

$$\text{MDD} = \max_t \left(\frac{\text{Peak}_t - \text{Trough}_t}{\text{Peak}_t} \right).$$

Calmar Ratio

The Calmar ratio relates annualized return to maximum drawdown:

$$\text{Calmar Ratio} = \frac{R_{\text{annual}}}{\text{MDD}}.$$

Beta

Beta measures portfolio sensitivity to market movements:

$$\beta = \frac{\text{Cov}(R_p, R_m)}{\text{Var}(R_m)},$$

where R_m denotes market returns.

Alpha

Alpha measures excess return relative to market exposure:

$$\alpha = E[R_p] - (R_f + \beta(E[R_m] - R_f)).$$

Treynor Ratio

The Treynor ratio measures excess return per unit of systematic risk:

$$\text{Treynor Ratio} = \frac{E[R_p - R_f]}{\beta}.$$

Why Multiple Metrics Are Necessary

Option strategies often exhibit skewed and non-normal return distributions. Metrics such as the Sharpe ratio may fail to capture downside risk accurately. Using a combination of metrics provides a more comprehensive evaluation of risk-adjusted performance.

Conclusion

Options allow market participants to reshape risk exposures in flexible and nonlinear ways. Evaluating option-based portfolios requires performance metrics that account for asymmetry, drawdowns, and market exposure. Together, option theory and risk metrics form the foundation of professional portfolio analysis.